

Møller Magnet Field Mapper

12 March 2009

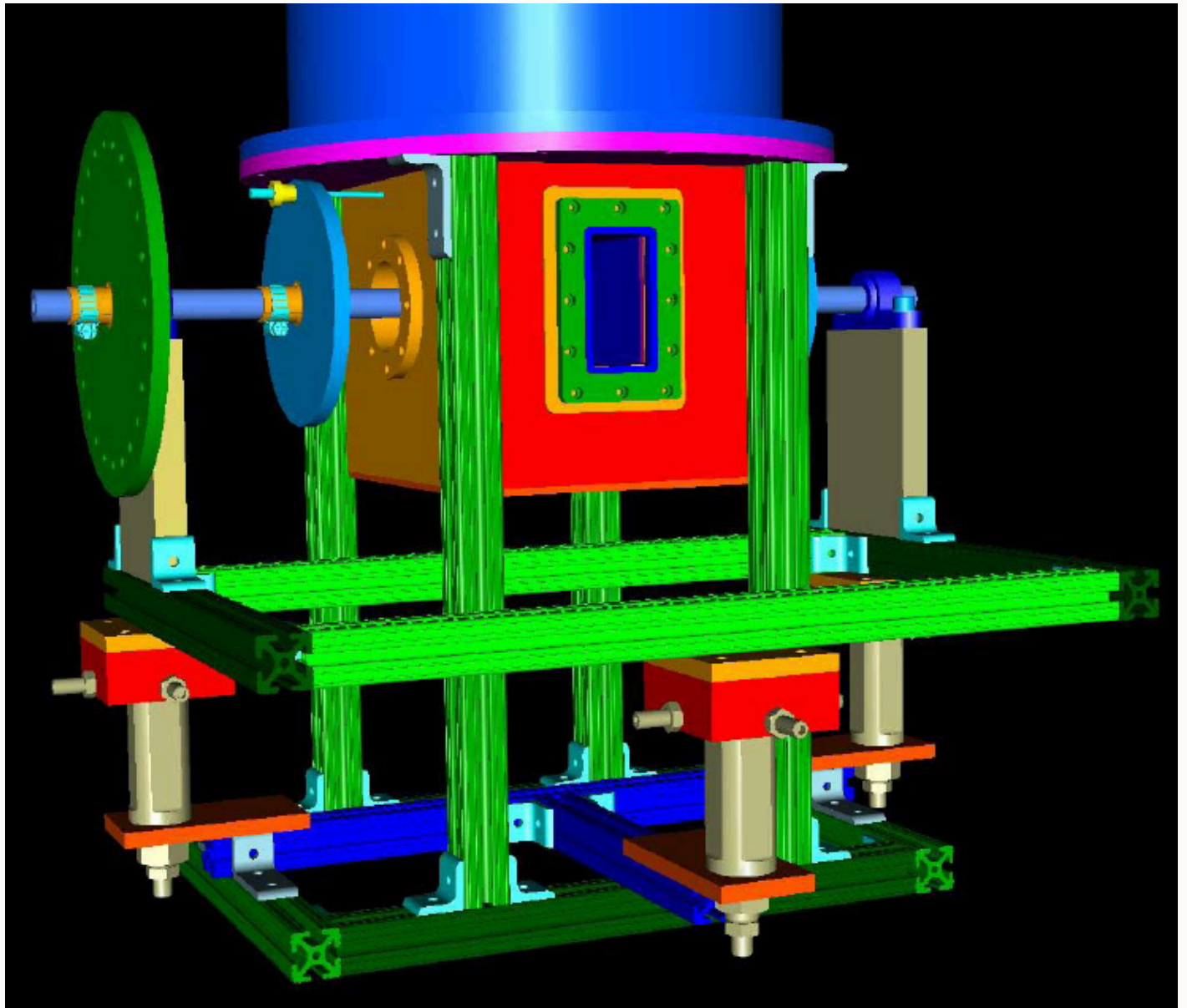
In June-July 2009 a new Møller target chamber with a superconducting magnet (4Tl field) has to be installed on the Hall A beam line. The magnetic field should be aligned along the beam line with the accuracy better than $\pm 0.5\text{mm}$ and $\pm 0.5\text{mrad}$. To perform alignment on the beam line, one has to know the magnetic field geometry inside the magnet chamber in accordance to some reference points on the chamber surface, i.e. the shift and inclination of the magnetic field axes in accordance to the magnet chamber axes. A magnetic field mapper should be used for this purpose.

A design of the field mapper is presented on the [picture](#). An axial (along z-axes) component of the magnetic field with the radius $\sim 9\text{cm}$ and on the distance $+14\text{cm}$ and -14cm from the target chamber center has to be measured (see the [picture](#)). Lake Shore axial probe [MMA-1802-VG](#) should be used for the measurements. The measurements procedure should be as follows:

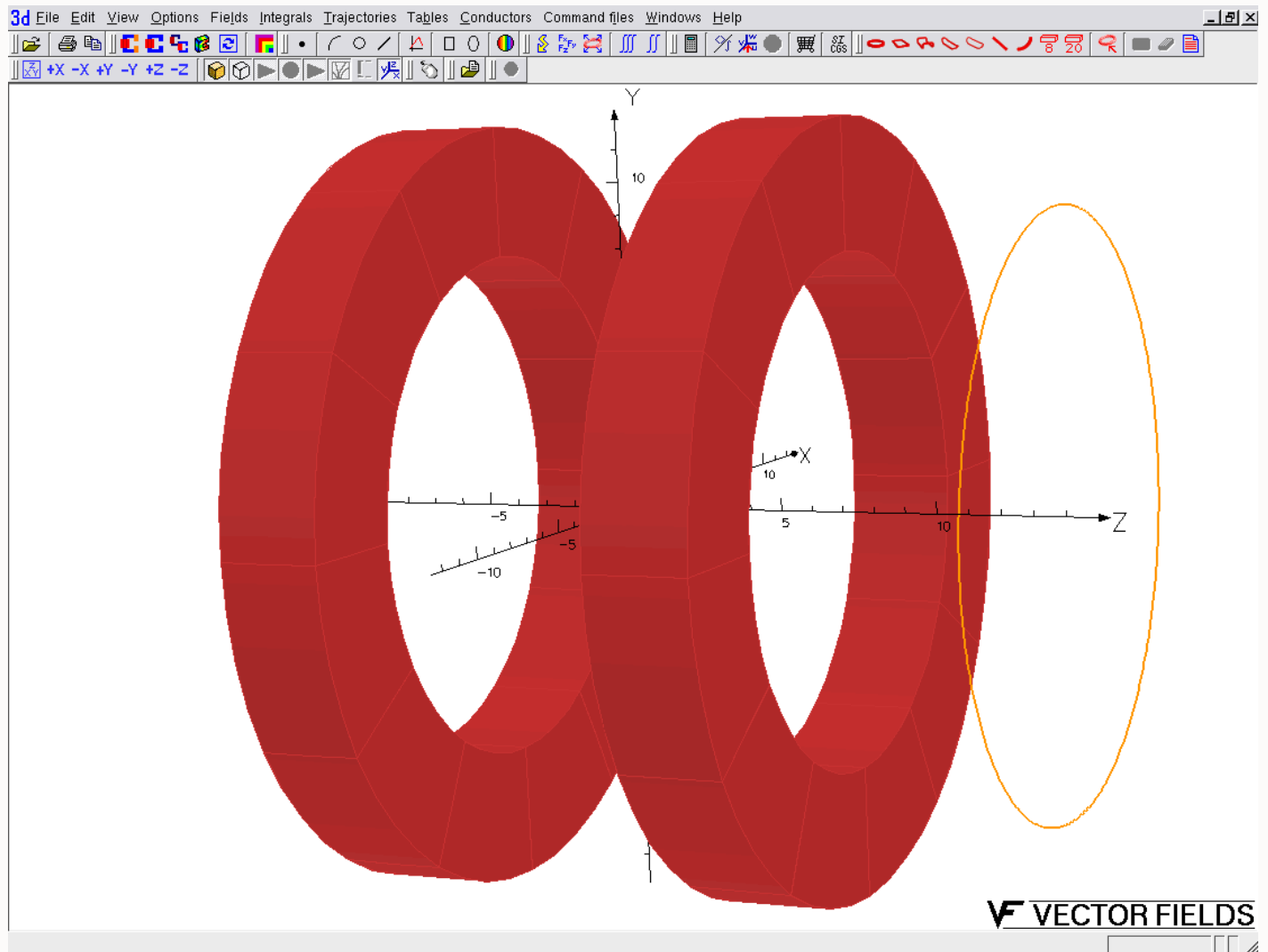
1. Calibrate the field probe.
2. Put the field probe into the field mapper socket.
3. Position the field mapper so that the distance between the field probe edge and the target chamber central plane (xy plane, $z=0$) is $+14\text{cm}$.
4. Perform alignment of the mapper ring axes with the target chamber axes.
5. Measure the axial (along z-axes) magnetic field component (36 points along the circle with radius 9cm).
6. Repeat 1-5.
7. Repeat 1-6 for the mapper position -14cm relative to the target chamber center.
8. Analyze data. Three components of the magnetic field, measured at the points on the circle with the radius 9cm lying in the central plane of the chamber, presented on the [plot](#). Measured results should look like those shown in [plot1](#) or [plot2](#). The black curves in the plots are the possible results of the measurements made at the position $+14\text{cm}$. The red curves are the possible results of the measurements made at the position -14cm . The green curves are the expected results with the ideal alignment of the magnetic field and the target chamber. An analysis of the field amplitude and phase for the two symmetrical field probe positions gives information about the magnetic field shifts and inclinations in respect to the field probe and to the reference points on the target chamber.
9. Repeat 1-7 with the field mapper position corrected in accordance with the analysis of the results. If the alignment of the field probe along the magnetic field axes is good enough, the measured result will be similar to the one shown in [plot3](#) or even more flat.

Figures

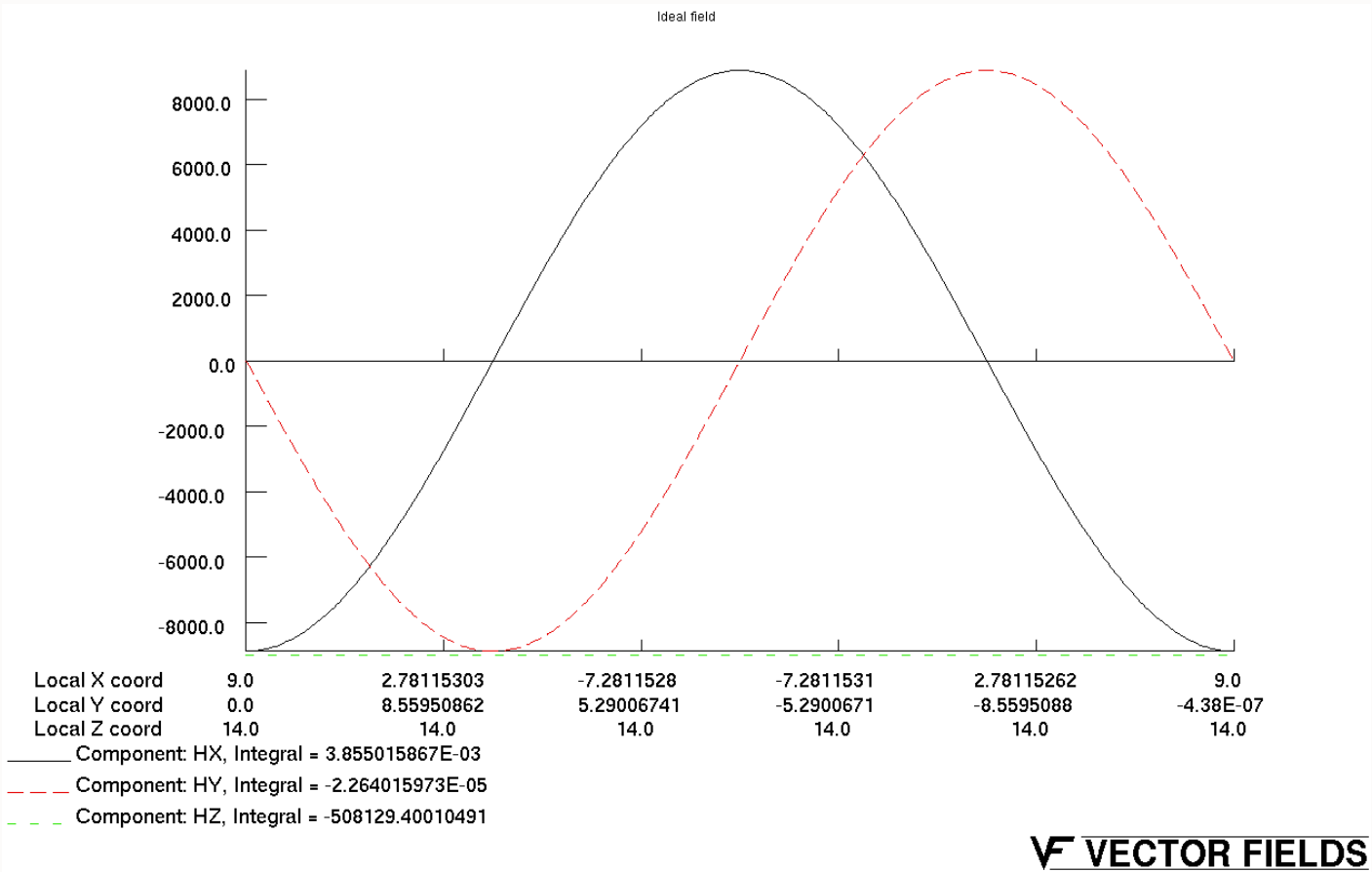
Field mapper design



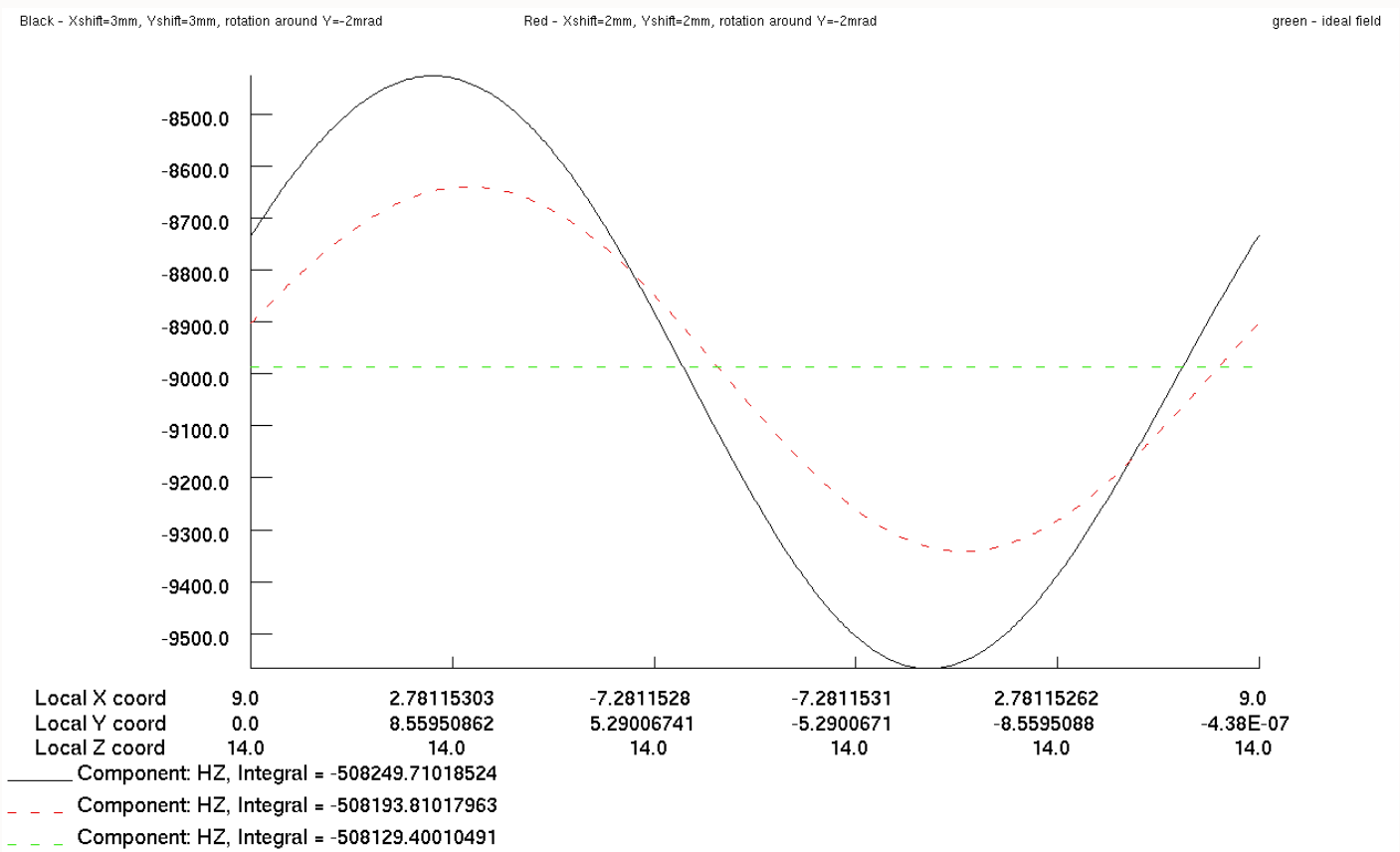
Axial field measurement geometry (~9 cm radius, ± 14 cm from chamber center)



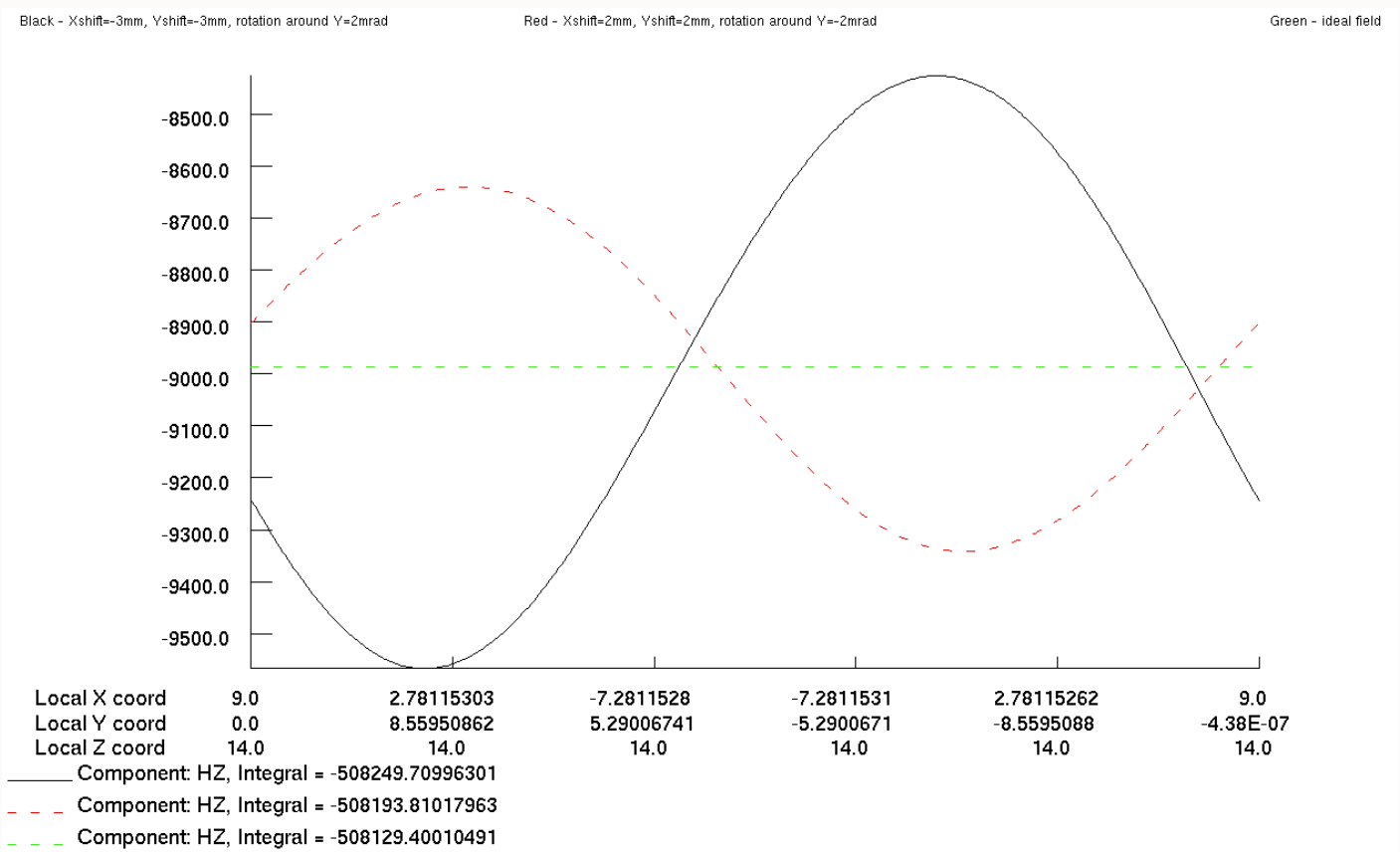
Three field components on the 9 cm circle in the chamber central plane



Plot1 — possible measured results at +14 cm (black) and -14 cm (red)

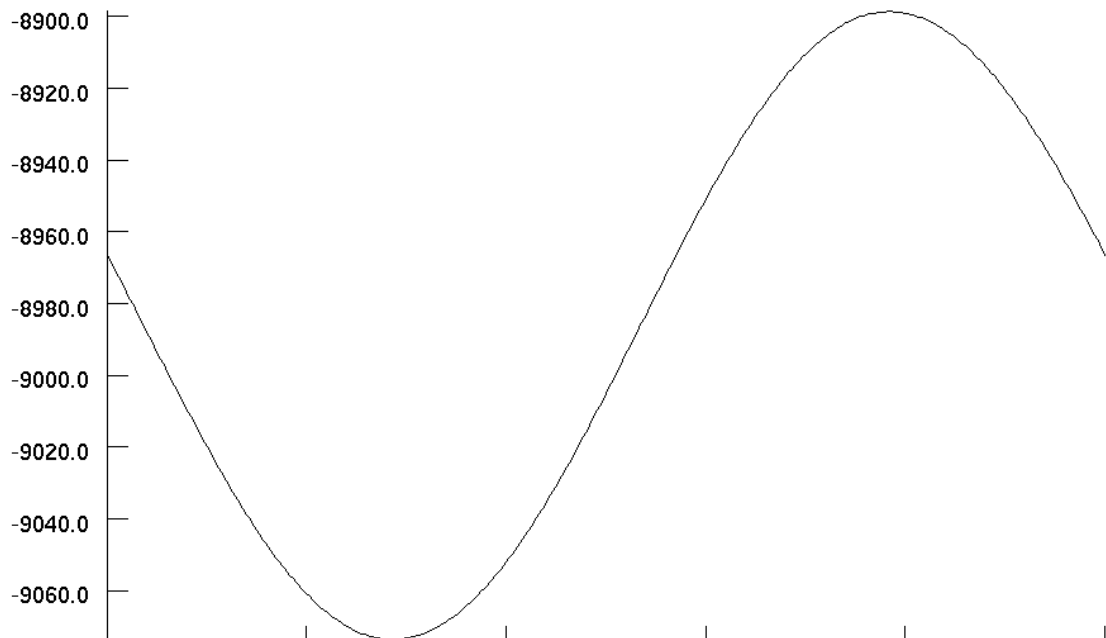


Plot2 — possible measured results at +14 cm (black) and -14 cm (red)



Plot3 — result with good alignment of the field probe along the magnetic field axes

Goal - Xshift=0.5mm, Yshift=-0.5mm, rotation around X=0.5mrad, around Y=-0.5mrad



Local X coord	9.0	2.78115303	-7.2811528	-7.2811531	2.78115262	9.0
Local Y coord	3.9031E-18	8.55950862	5.29006741	-5.2900671	-8.5595088	-4.38E-07
Local Z coord	14.0	14.0	14.0	14.0	14.0	14.0

Component: HZ, Integral = -508132.34852158